

# CoPianist: Co-Playing with a Virtual Pianist in an Interactive VR System for Piano Practice

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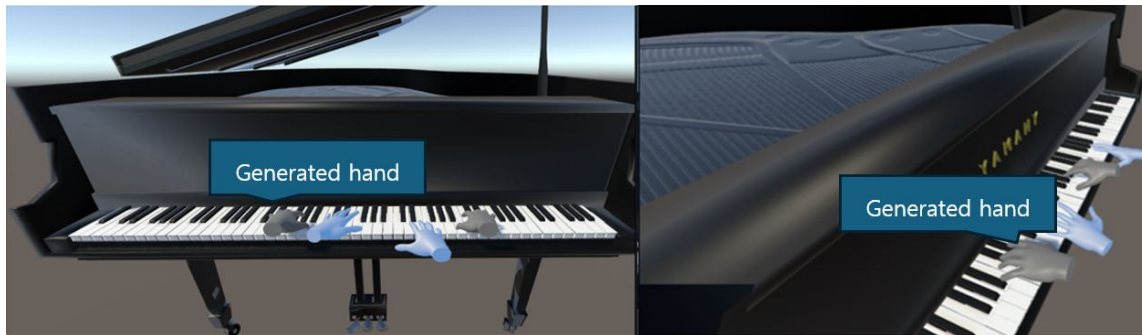


Figure 1: Visualization of multiview of CoPianist's generated virtual hands performing on a simulated piano

## ABSTRACT

Professional piano practice often lacks immediate and expressive feedback when conducted without an instructor or collaborative partner. This pilot study explores how immersive and generative technologies can enhance advanced piano training through CoPianist, a virtual reality system integrating generative AI for MIDI-based guidance and virtual hand simulation. The system combines adaptive MIDI performance generation with synchronized virtual hand movements, allowing learners to observe and co-perform with a virtual pianist on a simulated piano. Using hand tracking and real-time MIDI analysis, CoPianist delivers feedback on note accuracy, tempo stability, and technical execution.

**Index Terms:** Virtual reality, generative artificial intelligence, human-computer interaction, training and education

## 1 INTRODUCTION

Effective piano practice requires precise feedback on timing, accuracy, and technique, traditionally provided through one-to-one instruction or ensemble rehearsal. For piano professionals to be successful in performance, they need a high level of technical and emotional control, developed through sustained and focused practice [4]. However, professional and pre-professional pianists often practice alone, relying on self-monitoring and delayed feedback, which can limit both motivation and expressive refinement.

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While existing digital and MIDI-based tools provide objective performance analysis, they typically lack embodied interaction and the sense of co-presence found in human musical collaboration [1]. Virtual reality (VR) has the potential to improve the accessibility and scalability of piano practice in terms of cost, physical space, and learning efficiency, particularly for groups of learners. While VR equipment requires an initial investment, providing multiple physical pianos for group instruction entails substantially higher costs, space requirements, and limited facility availability. While existing digital and MIDI-based tools provide objective performance analysis, they often lack embodied interaction and the sense of co-presence found in human musical collaboration [1]. Recent advances in VR hand tracking and high-fidelity virtual instruments support embodied music interaction, while generative AI has shown promise in adaptive MIDI generation and expressive control. However, the integration of generative AI with embodied VR systems for advanced piano practice remains underexplored.

In this on-going research, we present a virtual reality piano practice system that incorporates generative AI to produce adaptive MIDI performances [2] and synchronized virtual hand simulations. The system simulates a grand piano within a fully interactive virtual environment, allowing users to practice alongside a virtual pianist that demonstrates musical passages and adapts to the learner's tempo and performance.

This work contributes the design of a VR piano practice system integrating generative AI for MIDI and hand performance simulation.

## 2 OVERALL SYSTEM DESIGN

### 2.1 System Architecture

CoPianist is an immersive piano practice system that integrates virtual reality, generative AI, and real-time performance analysis Fig. 2. Musical audio input is converted into temporally aligned symbolic (MIDI) and motion representations, which are rendered in a Unity-based VR environment where a co-present virtual pianist performs alongside the user. User performance is captured via hand tracking and evaluated using complementary metrics, including note and timing accuracy, hand positioning, and motion natu-

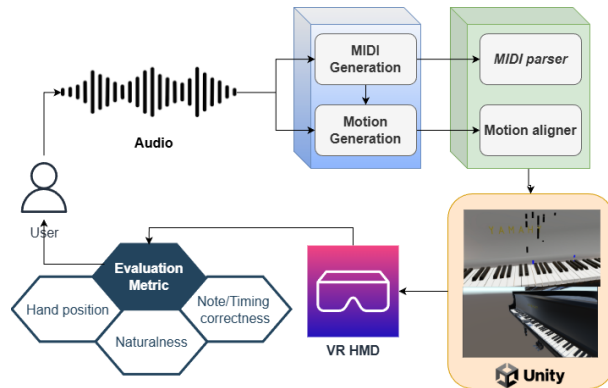


Figure 2: Overall framework of our system

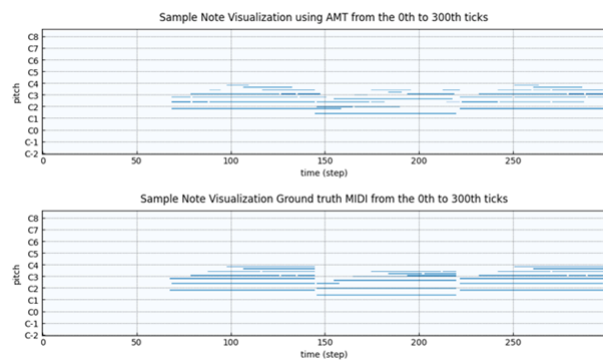


Figure 3: A visualization of a song segments from 0th to 300th ticks showing note activation over octaves

ralness, enabling both objective assessment and embodied practice feedback.

## 2.2 Audio-to-MIDI Generation and Note Representation

CoPianist employs a pitch-analysis neural network, Basic Pitch [2], to process audio input and estimate note onsets, pitches, and offsets. The model output is converted into a full MIDI file that serves as a symbolic representation of the reference performance. Fig. 3 illustrates a representative comparison between notes estimated from audio and ground-truth MIDI over the same temporal window. The generated full MIDI file is parsed by a MIDI parser module that extracts note-level attributes, including note identity, onset time, duration, and beat position. These attributes form a unified musical representation that is used consistently across visualization, virtual performance playback, and user performance analysis. While direct MIDI input would simplify the pipeline and improve transcription accuracy, it requires a MIDI keyboard and additional recording steps. This design choice prioritizes accessibility over transcription accuracy, while acknowledging that transcription errors such as slight onset shifts or fragmented note as shown in Fig. 3.

## 2.3 MIDI Visualization and Practice Customization

Extracted note data is rendered through a MIDI parser module within the virtual environment. Notes are visualized in relation to the virtual keyboard to support timing and pitch guidance during practice. Visualization parameters such as note color and playback speed can be adjusted, allowing users to customize practice difficulty and focus on specific musical elements. In the Unity environ-

ment, MIDI attributes are converted to seconds to improve synchronization with audio and timing.

## 2.4 Generative AI-Driven Virtual Pianist and Hand Simulation

To generate realistic and continuous piano hand motions, CoPianist builds on prior data-driven hand motion generation approaches [3]. The generative model is conditioned on musical audio and produces temporally aligned hand motion sequences, which are synchronized with audio and MIDI and streamed to a virtual pianist avatar as 26-joint skeletal per hand data for animation in Unity. While existing models demonstrate high-fidelity motion synthesis, they are primarily designed for offline generation and limited to short segments (approximately 8 seconds). In this work, we incorporate MIDI alongside audio to provide a more precise temporal context, improving note-level alignment and timing, and plan to extend the system to full-body performance generation, including pedal and upper-body motion, in future work.

## 2.5 User Interaction and Performance Analysis

User interaction in CoPianist is enabled through hand tracking, allowing pianists to perform directly on the virtual keyboard using natural hand and finger movements. The system captures user hand motion and key interactions in real time and aligns them with a reference performance.

Performance correctness is evaluated using MIDI-based analysis, which provides a discrete and timing-accurate representation of note events, enabling reliable assessment of note and timing accuracy independent of hand-tracking noise. Motion naturalness is evaluated separately by comparing the user's tracked hand motions with the generated virtual pianist's hand simulation, focusing on kinematic qualities such as hand positioning and movement continuity. By separating note-level correctness from motion assessment, CoPianist avoids conflating expressive movement quality with musical accuracy.

## 3 CONCLUSION

This work presented CoPianist, a virtual reality piano practice system that integrates generative AI, real-time MIDI analysis, and hand-tracked virtual performance to support advanced piano practice. By enabling learners to observe and co-perform with a virtual pianist while receiving immediate feedback on timing, accuracy, and technique, the system addresses key limitations of solitary professional practice. Future work will expand the system to support longer-term practice studies and full-body performance modeling, including pedal and posture analysis, to further explore the role of embodied virtual partners in professional music practice and combine with haptic gloves for haptic cues feedback.

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